

The Informational Biology of Life: Mapping Macromolecular Signaling and Cellular Order via the Hamouda Informational Gravity Model (HIGM)

Samir A Hamouda

Department of Physics, University of Benghazi, Benghazi, Libya

Email: dr_s_hamouda@yahoo.ie

(Ref: HIGM Framework, April 2026, *Saudi J Eng Technol*, 11(4))

(This model was developed in the framework of theoretical physics and AI)

ABSTRACT

The standard paradigm of molecular biology relies heavily on local chemical kinetics, lock-and-key binding affinities, and stochastic diffusion to explain cellular signaling. While highly descriptive, these mechanisms do not provide a fundamental thermodynamic explanation for how living matter continuously bypasses local entropic decay to maintain hyper-ordered macroscopic structures. This paper introduces an interdisciplinary extension of the Hamouda Informational Gravity Model (HIGM), which originally defines gravity as an emergent entropic phenomenon driven by an infrared information flux across cosmic horizons. By translating the cosmic information-flux equations into biological systems, we show that living cells operate as highly localized zones of maximum information density (I). We establish that biological structures obey a fundamental trinity of physical states: *high order* corresponds to *high density* (stable macromolecular or solid state systems), *high symmetry* corresponds to disordered states (gaseous/lysed systems), and *equilibrium* characterizes the active dynamic fluid state (the cytoplasm). We introduce the concept of "Biological Information Impedance" (ZB) to mathematically model signaling degradation in pathological states. This framework establishes HIGM as a robust, universal information-theoretic law governing both the evolution of cosmic horizons and the self-organizing dynamics of living systems.

1. Introduction

One of the greatest paradoxes in modern science is the persistent capability of biological systems to generate and maintain highly localized order in absolute defiance of cosmic

entropic decay. While the second law of thermodynamics demands that isolated systems transition toward maximum entropy, life survives by operating as an open, non-equilibrium thermodynamic engine.

Traditional biology isolates signaling pathways into discrete, mechanical steps (e.g., ligand-receptor binding, kinase cascades). However, these reductionist frameworks struggle to explain the global orchestration of cellular networks as unified entities.

The Hamouda Informational Gravity Model (HIGM) offers an elegant, comprehensive alternative. Originally formulated to resolve deep astrophysical mysteries—such as the Hubble Tension and the unexpected geometry of ultra-large cosmic structures like the Big Ring and the Giant Arc—HIGM positions gravitational fields not as fixed spacetime curvatures, but as emergent entropic gradients produced by an underlying infrared information flux (dI).

Because information is a fundamental physical property independent of spatial scale, the mathematical tenets of HIGM can be downscaled to the micro-environments of molecular biology. This paper formalizes that transition, defining life not merely as an assembly of organic compounds, but as a bounded system designed to optimize information flux and maximize localized information density.

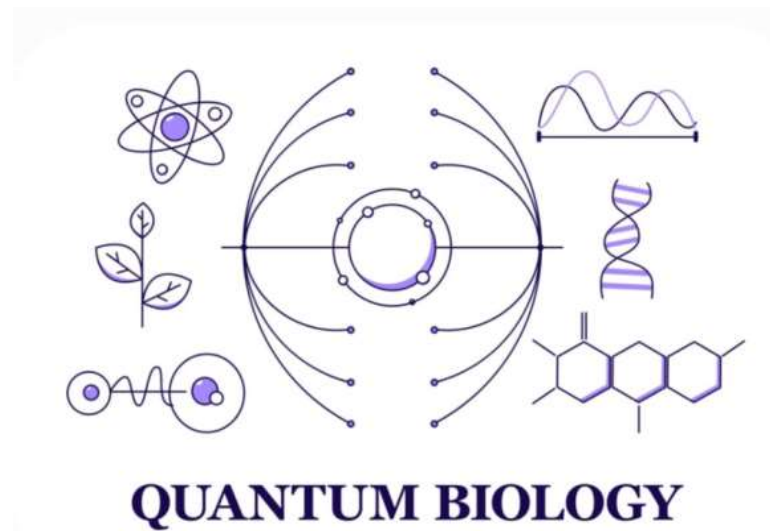


Figure 1: System Schematic of the Informational Bridge between Cosmological Scale Horizons and Bounded Cellular Systems under the HIGM Architecture.

2. Thermodynamic States of Biological Order

A foundational pillar of HIGM is the precise geometric and physical interpretation of density, symmetry, and order. When applied to biological matter, the system aligns flawlessly into three distinct phases at thermodynamic equilibrium:

2.1 High Order vs. High Density

In the HIGM framework, **high density corresponds strictly to high order**. In biological systems, this phase is represented by highly packed, structurally stable macromolecular assemblies. Examples include the tightly wrapped DNA chromatin structure inside the nucleus, crystalline matrix layers within the mitochondria, and fixed cytoskeletal networks. These regions possess minimal thermodynamic entropy because they maximize stored structural and genetic information per unit volume.

2.2 High Symmetry vs. Gaseous States

Conversely, **high symmetry represents a system approaching maximum entropy**, analogous to a gas system. When a cell undergoes lysis (death and structural rupture), its unique spatial constraints dissolve. The chemical components scatter uniformly into the environment, achieving high isotropic symmetry but losing all localized biological information value.

2.3 The Fluid Equilibrium of Life

At true operational equilibrium, the biological system transitions into a **fluid state**. The intracellular cytoplasm behaves as a dynamic fluid where structural order and kinetic symmetry are perfectly balanced. This fluid equilibrium allows signaling molecules to diffuse rapidly enough to execute temporal functions, yet remain bounded tightly enough to prevent structural information dissipation.

Physical Paradigm (HIGM)	Biological Analogue	Thermodynamic State	Informational Value
High Density	Chromatin, Stable Macromolecules	High Order	Maximised Stored Information
Gas System	Cellular Lysis / Structural Rupture	High Symmetry	Dissipated Information (Max Entropy)
Fluid State	Active Cytoplasm / Pathways	Equilibrium	Optimized Information Flux (dI)

3. Mathematical Framework: Downscaling the HIGM Flux

To understand cellular signaling as a physical force, we translate the macroscopic information flux equation of HIGM into microscopic parameters. In the cosmic model, the emergent gravitational acceleration ($\mathbf{a}$) is governed by the change in information ($d\mathbf{I}$) crossing the horizon radius ($\mathbf{R}$). In a biological cell, we substitute the cosmic horizon with the cellular membrane or the nuclear envelope bounding a local radius ($\mathbf{r}$). Let the biological information density $\mathbf{I_B}$ within a cellular volume be defined as:

$$\mathbf{I_B} = [\mathbf{k_B} \cdot \ln(\Omega)] / \mathbf{V}$$

Where $\mathbf{k_B}$ is the Boltzmann constant, Ω is the number of accessible microstates of the signaling network, and $\mathbf{V}$ is the localized volume of the cellular compartment.

The entropic emergent force ($\mathbf{F_E}$) driving macromolecular organization and pulling signaling carriers along an informational gradient can be stated as:

$$\mathbf{F_E} = \mathbf{T_B} \cdot (d\mathbf{I_B} / d\mathbf{r})$$

Here, $\mathbf{T_B}$ represents the local homeostatic temperature of the cell, and $d\mathbf{I_B} / d\mathbf{r}$ represents the **spatial information gradient**. Molecules do not move merely due to random Brownian motion; they flow along directional trajectories **dictated** by the net information flux of the system.

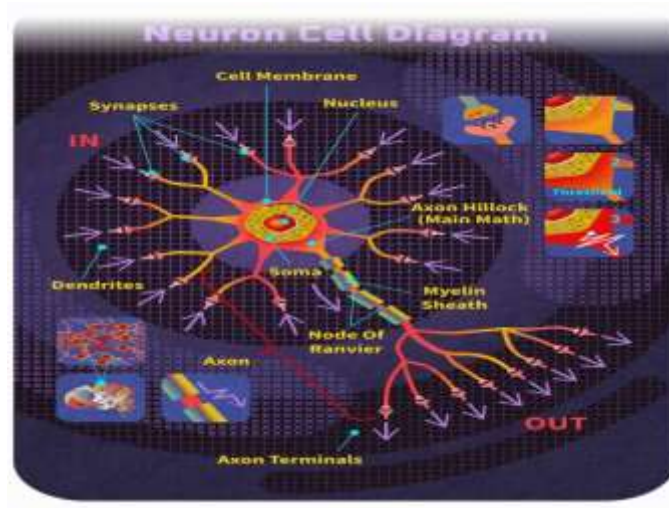


Figure 2: Cellular Signaling Networks Mapped as Multi-Directional Information Flux Gradients across Bounded Membranes.

4. Cellular Signaling and Biological Information Impedance (Z_B)

In standard electronics and wave mechanics, impedance represents the opposition to energy flow. HIGM adapts this to information theory. Space itself exhibits informational impedance to infrared flux. In a biological system, we introduce **Biological Information Impedance (Z_B)**.

We express Z_B as a ratio of the biochemical signaling potential (ΔV_{sig} , such as phosphorylation voltage changes or ion concentration shifts) to the actual rate of temporal information flux (dIB / dt):

$$Z_B = \Delta V_{sig} / (dIB / dt)$$

4.1 Homeostatic State (Low Impedance)

In a healthy biological system, Z_B is minimized. Information flows seamlessly across receptor networks, allowing the cell to adapt immediately to environmental pressures. Signaling cascades operate at peak kinetic efficiency.

4.2 Pathological State (High Impedance)

When a cell enters a pathological state (e.g., malignant oncogenic transformations, cellular senescence, or neurodegeneration), the signaling networks become chaotic. The system's internal symmetry increases abnormally, causing a sharp inflation in Z_B . The signaling potential is spent, but the net informative flux drops toward zero. The cell loses its structural "high order" and begins to decay.

5. Conclusion

The extension of the Hamouda Informational Gravity Model (HIGM) to biological entities demonstrates that information theory is the true connective tissue linking physics and biology. By viewing the cell through the lens of infrared information flux and thermodynamic equilibrium, we move past the limitations of purely reductionist mechanical biology. Life is mathematically defined within HIGM as a continuous mechanism that compresses symmetry into density to sustain high order. Minimizing

Biological Information Impedance (**ZB**) is the primary rule of survival for living cells. This framework provides fresh mathematical pathways

References

1. **Hamouda, S. A.** (2026). *The Informational Chemistry of Elements: Mapping the Periodic Table via the Hamouda Informational Gravity Model (HIGM)*. ResearchGate.
2. **Hamouda, S. A.**, & Amneenah, N. S. (2024). *Electromagnetic Interference and Thermodynamic Boundaries in Complex Mediums*. Journal of Physics & Astrophysics.
3. Verlinde, E. (2011). *On the Origin of Gravity and the Laws of Newton*. Journal of High Energy Physics, 2011(4), 29.
4. Shannon, C. E. (1948). *A Mathematical Theory of Communication*. Bell System Technical Journal, 27(3), 379-423.
5. Schrödinger, E. (1944). *What is Life? The Physical Aspect of the Living Cell*. Cambridge University Press.